INDIAN SPACE HISTORY



1962: The **Indian National Committee for Space Research (INCOSPAR)** is formed under the Department of Atomic Energy.

1964: NASA launches the **Syncom-3** satellite, facilitating the first live international television broadcast of the 1964 Tokyo Olympic Games.

1967: The **Experimental Satellite Communication Earth Station (ESCES)** becomes operational in Ahmedabad to train engineers and test satellite applications.

1969: The **Indian Space Research Organisation (ISRO)** officially established, superseding INCOSPAR.

1972 (June) : The Government of India establishes the **Space Commission and the Department of Space (DOS)** to oversee national space activities.

1972 (September): ISRO is formally brought under the administrative control of the **Department of Space (DOS).**

1975–1976: The **Satellite Instructional Television Experiment (SITE)** is conducted, utilizing satellites to broadcast educational television programs to rural Indian villages.

1977–1979: The **Satellite Telecommunication Experiments Project (STEP)** is launched as a joint initiative between ISRO and the P&T Department, using the Franco-German Symphonie satellite to test domestic telecommunication systems.

1994: The first fully successful flight of the Polar Satellite Launch Vehicle was the **PSLV-D2** (4 stage), conducted on October 15, 1994. The mission successfully injected the **IRS-P2** remote sensing satellite into its intended sun-synchronous orbit.

"Dr. V. Narayanan assumed the charge of Secretary, Department of Space, Chairman, Space Commission and Chairman, ISRO on 13th January, 2025."

SPACE ORBITS

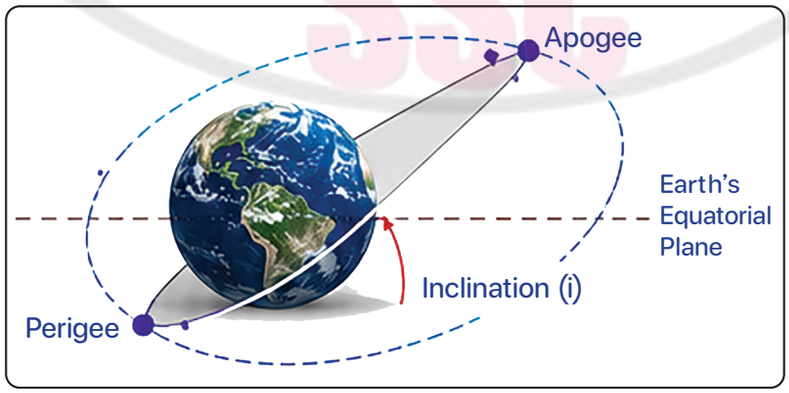
SPACE ORBITS



- **LEO (Low Earth Orbit):** Typically situated between 160 km and 2,000 km above Earth's surface. Used for the International Space Station (ISS) and most Earth observation satellites.
- **MEO (Medium Earth Orbit):** Located between LEO and GEO, typically around 20,000 km. Used primarily for navigation satellite systems like GPS.
- **GEO (Geostationary Orbit):** Located at approximately 35,786 km above the equator. The satellite moves in sync with Earth's rotation, appearing fixed to a specific point on the ground. Primarily used for telecommunications and weather satellites.
- **GTO (Geosynchronous Transfer Orbit):** An elliptical orbit used as an intermediate step to transfer a satellite from a low-altitude parking orbit into a higher geostationary orbit.
- **SSO (Sun-Synchronous Orbit):** A near-polar orbit that allows a satellite to pass over any given point on Earth at the same local solar time. Essential for imaging and weather monitoring.
- **HEO (Highly Elliptical Orbit):** An orbit with a very high eccentricity (an elongated ellipse). This allows the satellite to spend most of its orbital period over a specific high-latitude region.
- **Polar Orbit:** An orbit where the satellite passes above or near both poles of the planet. These are ideal for mapping and surveillance as the Earth rotates beneath the satellite's orbital plane.
- **Lagrangian Points (L1–L5):** Specific points in space where the gravitational pull of two large bodies (like the Earth and the Sun) equals the force required for a small object to move with them. Often used for deep-space observatories (e.g., Aditya-L1).

FUNDAMENTALS OF KEPLERIAN ORBITAL ELEMENTS

- **Perigee:** The point in an orbit closest to the Earth.
- **Apogee:** The point in an orbit farthest from the Earth.
- **Inclination (i):** The angle between the orbital plane of the satellite and the Earth's equatorial plane
  - **0° inclination:** Equatorial orbit (directly above the equator).
  - **90° inclination:** Polar orbit (passes over the North and South Poles).



LAUNCH VEHICLES

Introduction



India has two main operational launch vehicles for deploying spacecraft into orbit: the Polar Satellite Launch Vehicle **(PSLV)** and the Geosynchronous Satellite Launch Vehicle **(GSLV)**.

The Vikram Sarabhai Space Centre **(VSSC)** in Thiruvananthapuram designs and develops India's launch vehicles.

Liquid and cryogenic stages are developed by the Liquid Propulsion Systems Centre **(LPSC)** in **Valiamala** and the ISRO Propulsion Complex **(IPRC)** in **Mahendragiri.**

Integration and launching take place at India's spaceport, the Satish Dhawan Space Centre **(SDSC) SHAR**, which operates two launch pads for all PSLV and GSLV flights.

TYPES

RETIRED	OPERATIONAL	UNDER-DEVELOPMENT
• SLV-3 • ASLV	• PSLV • SSLV • GSLV • GSLV Mk III • Sounding Rockets	• HRLV • RLV-TD • Scramjet Engine-TD

DESCRIPTION

1 SATELLITE LAUNCH VEHICLE-3 (SLV-3)



- India's first experimental and indigenous launch vehicle used in 1980's Indian satellite.
- It was an all-solid, four-stage rocket standing 22 meters tall, weighing 17 tonnes, and capable of placing a 40 kg class payload into Low Earth Orbit (LEO).

2 AUGMENTED SATELLITE LAUNCH VEHICLE (ASLV)

- The ASLV program tripled SLV-3's capacity, aiming to orbit 150 kg class satellites into 400 km circular Low Earth Orbits (LEO).
- Standing 24 meters tall with a 40-tonne liftoff weight, ASLV was configured as a five-stage, all-solid propellant vehicle.
- First successful flight in **1992** (ASLV-D3).

3 POLAR SATELLITE LAUNCH VEHICLE (PSLV)

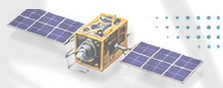


- The PSLV is India's third-generation launcher and its first to feature liquid stages.
- Since its maiden successful flight in October 1994, PSLV has served as India's highly reliable and versatile "workhorse" launch vehicle.
- **Configuration:**
  - Gross Weight (PSLV-XL): 320 tonnes
  - 4 stage launch vehicle
  - Variants: 4 (PSLV,CA,DL,QL,XL)
  - Payload to Sun Synchronous Polar Orbit: 1,750 kg
  - Payload to Sub-Geosynchronous Transfer Orbit: 1,425 kg
- Latest PSLV launches:
  - PSLV-C62/EOS-N1
  - PSLV-C61/EOS-09
  - PSLV-C60/SPADEX
  - PSLV-C59/Proba-3
  - PSLV-C58/XPoSAT
  - PSLV-C57/Aditya-L1

4 GEOSYNCHRONOUS SATELLITE LAUNCH VEHICLE (GSLV)

- The GSLV Mk II is India's launch vehicle designed to deliver communication satellites into Geostationary Transfer Orbit (GTO).
- While early flights relied on Russian-supplied cryogenic third stages, India transitioned to its own indigenously developed cryogenic engine starting with the successful GSLV D5 mission in January 2014.
- First successful operational flight in 2004 (GSLV-F01).
- Configuration:
  - 3 stage launch vehicle
  - Lift-off mass: 420 tonnes
  - Payload to GTO: 2,250 kg
  - Payload to LEO: 6,000 kg
- Latest PSLV launches:
  - GSLV-F16/NISAR
  - GSLV-F15/NVS-02
  - GSLV-F14/INSAT-3DS
  - GSLV-F12/NVS-01





5 GSLV MK III (LVM3)



- The **LVM3 (Launch Vehicle Mark-3)**, formerly known as the **GSLV Mk III,** is India's heaviest and most powerful operational launch vehicle.
- It features two solid strap-on motors (S200), a liquid core stage (L110), and a high-thrust cryogenic upper stage (C25).
- Powered by the fully indigenous CE20 high-thrust engine, the C25 cryogenic upper stage carries a propellant loading of 28 tons.
- **Configuration:**
  - 3 stage
  - Lift-off mass: 640 tonnes
  - Payload to GTO: 4,000 kg
  - Payload to LEO: 8,000 kg
- **Latest LVM3 launches:**
  - LVM3-M6/BlueBird Block-2
  - LVM3-M5/CMS-03
  - LVM3-M4-Chandrayaan-3
  - GSLV-Mk III - M1/Chandrayaan-2
- The **S200** is a solid rocket booster developed by the Indian Space Research Organisation (ISRO). It serves as the primary stage (first stage) for the **LVM3** (Launch Vehicle Mark-3).

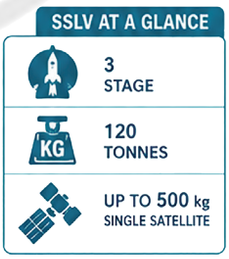




6 SMALL SATELLITE LAUNCH VEHICLE (SSLV)



- 3 stage Launch Vehicle configured with three Solid Propulsion Stages and liquid propulsion based Velocity Trimming Module (VTM) as a terminal stage.
- **Configuration:**
  - **Lift-off mass:** 120 tonnes
  - **Payload Capabilities (500 km Planar Orbit):**
  - **Single Satellite:** Up to 500 kg
  - **Multiple Satellites:** 10 kg to 300 kg each



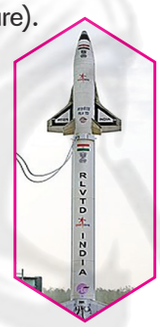


7 SOUNDING ROCKETS



- Sounding rockets are one- or two-stage solid propellant vehicles used to probe the upper atmosphere for space research.
- The establishment of the Thumba Equatorial Rocket Launching Station in 1963, strategically located near the magnetic equator, significantly advanced India's atmospheric and aeronomy research.
- This initiative officially launched the Indian space program with the successful firing of the first sounding rocket from Thumba, Kerala, on November 21, 1963.
- The first rockets were two-stage rockets imported from Russia (M-100) and France (Centaure).

8 REUSABLE LAUNCH VEHICLE – TECHNOLOGY DEMONSTRATOR (RLV-TD)



- ISRO's RLV-TD develops critical technologies for low-cost space access.
- The vehicle features a fuselage, nose cap, double delta wings, twin vertical tails, and active control surfaces.
- During testing, a low-burn-rate HS9 solid booster accelerated the craft to Mach 5.
- ISRO successfully flight-tested RLV-TD on 23 May 2016 from SDSC SHAR, Sriharikota.

9 SCRAMJET ENGINE-TD



- ISRO's scramjet engine uses hydrogen as fuel and extracts oxygen directly from the atmospheric air as its oxidizer.
- ISRO successfully conducted its first experimental scramjet engine test on August 28, 2016, from Sriharikota's Satish Dhawan Space Centre.
- Advancing air-breathing propulsion technology, the vehicle flew for approximately 300 seconds before splashing down in the Bay of Bengal, 320 km downrange.

10 HRLV (HUMAN RATED LAUNCH VEHICLE)



- The HRLV is the human-rated version of ISRO's LVM3 (Launch Vehicle Mark 3), specifically modified to ensure the safety of crew members for India's Gaganyaan (human spaceflight) mission.
- It is the vehicle tasked with launching Indian astronauts into Low Earth Orbit (LEO).

11 NEXT GENERATION LAUNCH VEHICLE (NGLV)

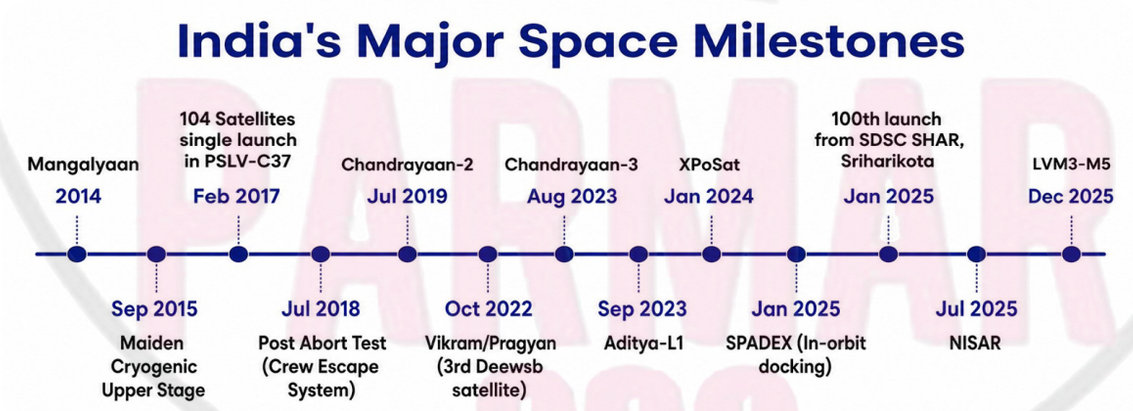


- The Next Generation Launch Vehicle (NGLV) is ISRO's upcoming heavy-lift, partially reusable launch vehicle family, designed to replace ageing systems like PSLV and supplement/supersede GSLV/LVM3 for future national and commercial missions.
- Three-stage, partially reusable launch vehicle family.
- Up to 30 tonnes to Low Earth Orbit (LEO) in the largest configuration.
- Around 10 tonnes to Geostationary Transfer Orbit (GTO).
- This is roughly 3 times the present payload capability compared to LVM3, with about 1.5 times the cost.
- Semi-cryogenic stages using refined kerosene (fuel) + liquid oxygen (LOX, oxidiser) for booster stages.
- Total approved fund: ₹8,240 crore

INDIAN SATELLITES

- **Aryabhata (1975):** India's first satellite, launched on 19 April 1975 from Kapustin Yar (USSR) using a Soviet rocket. Russia also enabled India's first human spaceflight milestone when Wing Commander Rakesh Sharma travelled to space aboard Soyuz T-11 in 1984.
- **Bhaskara-I and Bhaskara-II (1979, 1981):** First experimental remote sensing satellites built in India, dedicated to Earth observation and hydrology.
- **Rohini Series (1979–1983):** Rohini Technology Payload (1979) and Rohini RS-1 (1980) were the first satellites launched by an Indian-built launch vehicle (SLV-3).
- **INSAT Series:** INSAT-1B (1983) and subsequent INSAT satellites established India's national satellite communication and broadcasting system.
- **GSAT Series:** Advanced communication satellites like GSAT-8, GSAT-11, GSAT-29, and GSAT-31 increased high-throughput capacity for broadband, DTH, and defense communications.
- **Cartosat and IRS Series:** CARTOSAT-1 (2005) and later Cartosat-2 series (2007 onwards) provided high-resolution imagery for urban planning, agriculture, and resource management. IRS (Indian Remote Sensing) series was started with **IRS-1A** in **1988**.
- **RISAT, HysIS, and Microsat**
  - **RISAT-1** (2012) is India's first radar imaging satellite, enabling all-weather, day-night Earth observation.
  - **HysIS** (2018) is a hyperspectral imaging satellite for detailed land, water, and vegetation studies.
  - **Microsat-R** (2019) and **EMISAT** (2019) support defense imaging and electromagnetic intelligence, respectively.
- **Regional Navigation Satellite System (NavIC):** India's own regional navigation system, IRNSS (Indian Regional Navigation Satellite System), now called NavIC, began with IRNSS-1A (2013).
- **2017: PSLV-XL** launched **104 satellites** in a single mission – largest number of satellites launched by a single rocket.

INDIAN SPACE MISSIONS



GSAT-7R (CMS-03)

- India's launch vehicle successfully launched the CMS-03 multiband communication satellite in its 5th operational flight (LVM3-M5) on November 02, 2025.
- Launching station: Satish Dhawan Space Centre, Sriharikota
- The satellite is India's heaviest communication satellite till date, weighing approximately 4,400 kg, launched to Geosynchronous Transfer Orbit (GTO).
- Primary beneficiary: Indian Navy
- It is a replacement for the aging GSAT-7 (Rukmini) satellite launched in 2013.

BlueBird Block-2 Mission

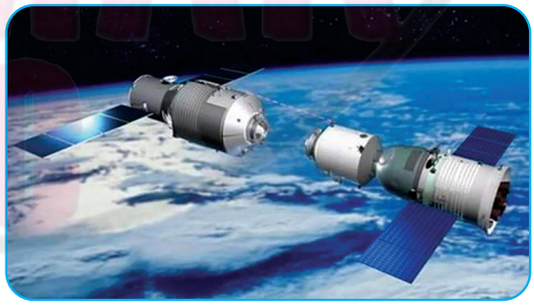


- Indian Space Research Organisation (ISRO) launched a next-generation US communication satellite BlueBird Block-2 onboard its heaviest vehicle LVM3-M6 from Sriharikota in Andhra Pradesh.
- It is part of a global LEO (Low Earth Orbit) constellation to provide direct-to-mobile connectivity through satellite.
- Mass (Weight): 6,100 kg per satellite
- The mission was undertaken as part of the commercial agreement signed between NewSpace India Ltd (NSIL) and US-based AST SpaceMobile.

India's Space Vision 2047

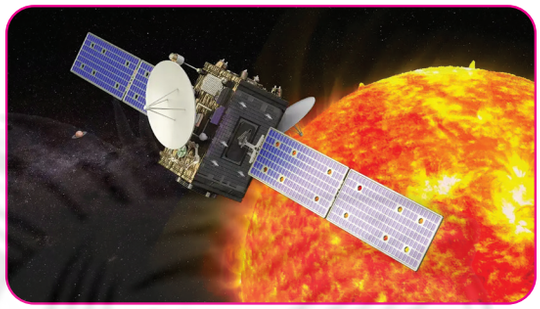
Achievement	Target Year
Establishment of BAS 1st module	2028
Bharatiya Antariksh Station (BAS)	2035
Landing of an Indian on Moon	2040
Development of Next Generation Satellite Launch Vehicle (NGLV)	2032
Chandrayaan-4	2027-28
Venus Orbiter Mission (VOM)	2028

SpaDeX (Space Docking Experiment)



- The SpaDeX (Space Docking Experiment) mission is ISRO's cost-effective technology demonstrator to validate in-space rendezvous, docking, and undocking using two small spacecraft, **SDX01 (Chaser) and SDX02 (Target)**, each weighing about 220 kg, launched into a **470 km low-Earth orbit** at 55° inclination.
- **Launch:** December 30, 2024, **PSLV-C60** deploys satellites with initial 10-20 km separation; drift arrest achieves far rendezvous.
- **Docking:** **January 16, 2025,** precise maneuvers from 15 m to capture, retraction, rigidization, and integrated control.
- **Payload:** PS4-Orbital Experiment Module, designated as **POEM**, refers to the usage of the spent fourth stage of the Polar Synchronous Launch Vehicle (PSLV).
- **India** became the **4th country** to do so after the USA, Russia and China.

Aditya L1 Mission



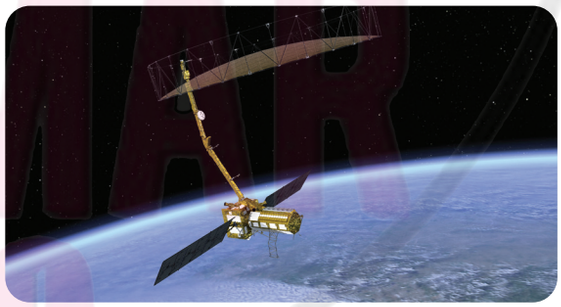
- Aditya-L1 is India's first space-based solar observatory mission, launched by ISRO on **September 2, 2023**, using **PSLV-C57** from Satish Dhawan Space Centre. The 1,500 kg spacecraft carries seven indigenously developed payloads to study the Sun's photosphere, chromosphere, corona, solar winds, and space weather dynamics. Positioned in a halo orbit around the **Sun-Earth L1 Lagrange point, 1.5 million km from Earth.**
- **Payloads:** VELC, SUIT, SoLEXS, HEL1OS, ASPEX, PAPA and a Magnetometer.
- Aditya-L1 reached L1 on **January 6, 2024**, after a **110-day journey** involving multiple Earth-bound orbits. It has captured solar flares, Coronal Mass Ejections (CMEs), high-energy X-rays, and full-disk UV images during transit. **SUIT detected solar flare 'kernels'** linking plasma heating, while VELC observed CMEs; the mission prepares for peak solar activity in 2026.

Axiom Mission 4



- Axiom Mission 4 (Ax-4) was a private crewed spaceflight to the International Space Station (ISS), operated by Axiom Space with SpaceX and NASA partners. It was launched on **June 25, 2025,** from **Kennedy Space Center.**
- **Commander:** Peggy Whitson (Axiom Space, USA; fifth spaceflight).
- **Pilot: Shubhanshu Shukla** (ISRO, India; first spaceflight).
- **Other crew members:** Slawosz Uznanski-Wisniewski (Poland) & Tibor Kapu (Hungary)
- **Rocket:** Falcon 9
- The crew conducted over 60 experiments from 31 countries in microgravity, focusing on human physiology, Earth observation, life sciences, and materials.
- It tested **Crew Dragon Grace,** the final capsule built, and supported future missions like lunar/Mars exploration.

NISAR Mission

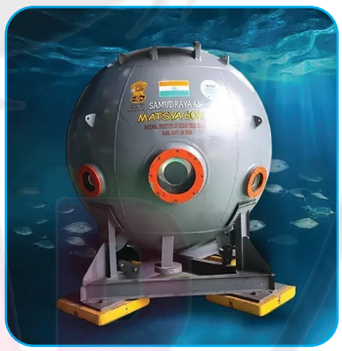


- The NASA-ISRO Synthetic Aperture Radar (NISAR) mission is a historic joint satellite project between **NASA** and the Indian Space Research Organisation **(ISRO)**, launched on **July 30, 2025**, from ISRO's **Satish Dhawan Space Centre.**
- It is the first Earth observation satellite to carry a dual-frequency radar system, featuring L-band (24 cm wavelength) and S-band (12 cm wavelength) **Synthetic Aperture Radar (SAR)** instruments.
- Weight: 2392 kg
- Rocket launcher: GSLV Mk II (GSLV-F16)
- NISAR operates in a **sun-synchronous orbit** at an altitude of **747 km** with an **orbital period of 100 minutes (12 day repeat cycle).**
- The mission's key objectives include measuring and monitoring changes in ecosystems, land deformation, ice masses, sea level rise, natural hazards (earthquakes, volcanoes, landslides), soil moisture, and groundwater resources.
- The NISAR spacecraft is built around the I-3K Structure (Lift-off mass: 2400 kg), which serves as the heritage spacecraft bus.
- Mission Life: 5 Years

Some Components of NISAR		
Name	**Details**	**Space agency associated**
L-Band SAR	Frequency - 1.25 GHz	NASA
S-Band SAR	Frequency - 3.2 GHz	ISRO
12 metre deployable mesh reflector antenna		NASA
9 metre deployable boom		NASA
High speed downlink system		ISRO

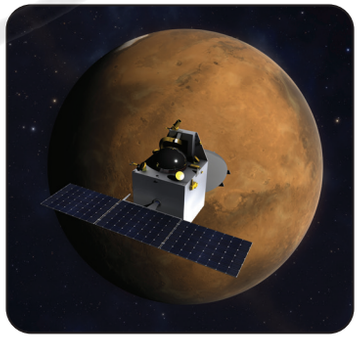
NISAR LAUNCHES		
Aspect	**100th Launch (GSLV-F15/NVS-02)**	**101st Launch (PSLV-C61/EOS-09)**
Date	Jan 29, 2025	May 18, 2025
Launch Vehicle	**GSLV Mk II**	**PSLV**
Payload	**NVS-02 (NavIC navigation)**	**EOS-09 (SAR Earth observation)**
Orbit	**GTO (Geosynchronous Transfer Orbit)**	SSPO (Sun Synchronous Polar Orbit)
Lift-off mass	**420 tonnes**	321 tonnes
Launch mass	2250 kg	1696 kg

Samudrayaan Mission



- Samudrayaan Mission is aimed to develop a self propelled manned submersible to carry **3 human beings** to a water depth of **6000 meters** in the ocean with a suite of scientific sensors and tools for deep ocean exploration.
- It has an endurance of **12 hours** of operational period and **96 hours** in case of emergency.
- The projected timeline is five years for the period **2020-2021 to 2025-2026.**
- **National Institute of Ocean Technology (NIOT), Chennai,** an autonomous institute under MoES, has developed a 6000m depth rated Remotely Operated Vehicle (ROV), named **ROSUB-6000.**
- India is set to launch its first human underwater submersible **MATSYA 6000** as a part of deep ocean mission (Samudrayaan).

Mars Orbiter Mission



- India's Mars Orbiter Mission (MOM), known as Mangalyaan, launched on **November 5, 2013,** via **PSLV-C25** from Sriharikota and entered Mars orbit on **September 24, 2014,** after a **298-day journey,** marking ISRO's first interplanetary success on the initial attempt.
- Originally designed for a six-month mission, Mangalyaan remained operational for more than eight years, far exceeding expectations.
- **Payloads :**
  - **Mars Colour Camera (MCC)**
  - **Methane Sensor for Mars (MSM)**
  - **Thermal Infrared Imaging Spectrometer (TIS)**
  - **Mars Exospheric Neutral Composition Analyser (MENCA)**
  - **Lyman Alpha Photometer (LAP)**

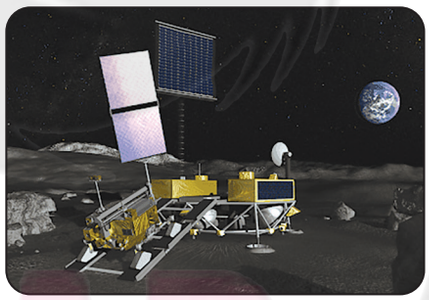
Mars Lander Mission (MLM) - Mangalyaan 2

- **Primary Objective:** To achieve a soft landing on Mars and deploy a rover and an aerial vehicle to conduct in-situ scientific research.
- The mission received formal approval from the Space Commission on **February 21, 2025.**
- It is currently in the development phase, with completion targeted for the **2028–2030** timeframe.
- Total spacecraft mass is estimated at around **4,500 kg,** to be launched by an **LVM3** (GSLV Mk III) rocket.
- The revised mission (MLM) will consist of **two main modules:**
  - Cruise Stage (Orbiter Module)
  - Descent Stage (Lander Module)
- **Orbiting Payloads:**
  - Mars Orbit Dust Experiment (MODEX)
  - Radio Occultation (RO) Experiment
  - Energetic Ion Spectrometer (EIS)
  - Langmuir Probe and Electric Field Experiment (LPEX)
  - PRISM (PolaRisation sensitive Infrared Spectroscopy of Mars)
  - PREM (Plasma AnalyseR for the Environment of Mars)



LUPEX Mission

- The Lunar Polar Exploration (LUPEX) mission, also known as Chandrayaan-5, is a collaborative effort between **ISRO (India) and JAXA (Japan)** to investigate water resources in the Moon's south polar region.
- ISRO provides the lander and select instruments, while JAXA supplies the 350 kg rover, **H3-24L launch vehicle,** and additional instruments; contributions come from NASA and ESA.



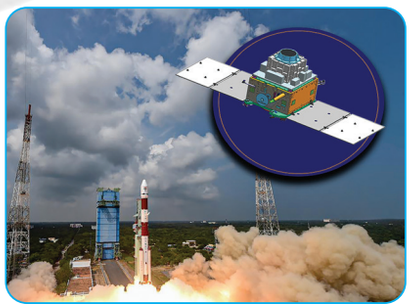
AstroSat

- India's Polar Satellite Launch Vehicle (PSLV-C30) successfully launched the AstroSat multi-wavelength observatory in its 31st flight on September 28, 2015.
- Satellite Details: The satellite is India's first dedicated space observatory, weighing approximately 1,513 kg, launched into a 650 km near-equatorial Low Earth Orbit (LEO).
- It is designed to simultaneously study celestial objects in a wide range of the electromagnetic spectrum, from visible light and ultraviolet to X-rays, making it a unique space-based observatory.



XPoSat (X-ray Polarimeter Satellite)

- India's Polar Satellite Launch Vehicle (PSLV-C58) successfully launched the XPoSat satellite in its 60th flight on January 1, 2024.
- The satellite is India's first dedicated polarimetry observatory, weighing approximately 469 kg, launched into Low Earth Orbit (LEO).
- It is India's first, and only the world's second, dedicated mission to study the polarization of X-rays from celestial sources.
- It aims to investigate the physics of extreme objects like black holes and neutron stars by adding a new dimension to X-ray astronomy.



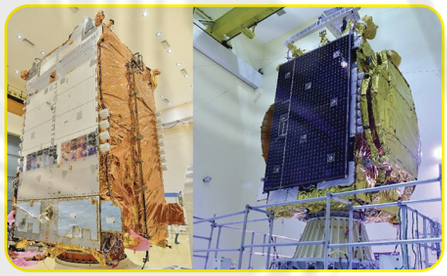
- Mission Objectives
  - POLIX (Polarimeter Instrument in X-rays) Payload: Measures the polarisation of X-rays (8-30 keV) from ~50 cosmic sources using Thomson scattering.
  - XSPECT (X-ray Spectroscopy and Timing) Payload: Conducts long-term spectral and temporal studies of X-ray sources (0.8-15 keV).

PSLV-C62 Mission



- Launch location: First Launch Pad at Satish Dhawan Space Centre, Sriharikota.
- Primary payload: Earth Observation Satellite - N1 (EOS-N1) also named as 'Anvesha'.
- Characteristics of PSLV-C62 Vehicle:
- Lift off Mass: 260 tonnes
- Stages: 4
- The PSLV-C62 mission encountered an anomaly during the end of the PS3 stage.

GSAT-N2 (also known as GSAT-20)



- High-throughput communication satellite developed by the Indian Space Research Organisation (ISRO) for **NewSpace India Limited (NSIL).**
- It operates primarily in Ka-Ka band frequency band to provide broadband and in-flight connectivity.
- **Launch Vehicle:** SpaceX **Falcon 9**
- **Mission Life:** 14 years.
- **Orbital Slot:** Geostationary Orbit (GSO) at 68° East longitude.
- **Mass:** 4,700 kg
- **Bus:** Built on the standard **I4K bus** (Carbon Fiber Reinforced Polymer structure).

SSLV-D3 Mission



- The **SSLV-D3** mission was the third and final developmental flight of ISRO's Small Satellite Launch Vehicle (SSLV).
- **Launch date:** August 16, 2024
- **Primary Payload:** EOS-08, an Earth Observation Satellite designed for microsatellite-based research.
- **Secondary Payload:** SR-0 DEMOSAT, a passenger satellite developed by Space Kidz India.
- **Orbit:** Both satellites were successfully injected into a circular Low Earth Orbit (LEO) at an altitude of 475 km with an inclination of 37.4°.

RLV-LEX Series

- **RLV LEX** (Reusable Launch Vehicle – Landing Experiment) is ISRO's series of test flights to demonstrate autonomous runway landing of a winged space vehicle.
- Purpose: To validate technologies for a winged, reusable launch vehicle that can:
  - Deliver satellites to orbit
  - Return from space
  - Glide and land horizontally on a runway like an aircraft, rather than vertically like Falcon 9.



- The vehicle used in LEX tests is called **"Pushpak".**
- An **Indian Air Force Chinook helicopter** carries the RLV as an underslung load.

- **RLV-LEX 03 details:**
  - Launch date: 23 June 2024
  - The **RLV-LEX-03** (Reusable Launch Vehicle Landing Experiment-03) was the third and final test in ISRO's series of landing experiments for its reusable space plane, **Pushpak.**
  - The test pushed the boundaries by releasing the vehicle with a **500-meter cross-range deviation** (compared to 150m in LEX-02) under more severe wind conditions, which the vehicle autonomously corrected to perform a precise centerline landing.

GSLV-F14/INSAT-3DS Mission



- The mission was ISRO's launch of India's third-generation meteorological satellite, **INSAT-3DS,** using the **Geosynchronous Satellite Launch Vehicle (GSLV)** to improve weather forecasting, environmental monitoring, and disaster warning capabilities.
- **Launch date:** 17 February 2024
- **Launch site:** Satish Dhawan Space Centre (SDSC-SHAR), Sriharikota, from the Second Launch Pad.
- **Launch vehicle:** GSLV-F14, a three-stage GSLV Mk.II with an indigenous cryogenic upper stage (CE-7.5).
- **Mission type:** Meteorological satellite deployment into Geosynchronous Transfer Orbit (GTO), followed by orbit-raising to place the satellite in a geostationary orbit at about 74° E longitude, 35,786 km altitude.
- **INSAT-3DS** was successfully separated about 19 minutes after launch and later stabilized in its operational geostationary slot.
- **Primary payloads:** Imager, Sounder, Data Relay Transponder (DRT) and Satellite-Aided Search and Rescue (SAS&R) transponder.
- **Structure:** I-2k platform

Gaganyaan



- Gaganyaan is **India's first human spaceflight program** by ISRO, aiming to launch a crew of **three astronauts** to a 400 km **low Earth orbit** for a **three-day mission.**
- The mission demonstrates indigenous human spaceflight capability using the human-rated **LVM3 (HLVM3)** launch vehicle.
- ISRO conducted its first **Integrated Air Drop Test (IADT-1)** for the Gaganyaan mission on **Aug. 24, 2025.**
- In October 2024, the scope of Gaganyaan programme was revised from three missions (G1, G2, H1) to eight missions. It includes three uncrewed missions G1, G2, G3 followed by two crewed missions (H1 & H2) followed by a docking mission to any space station (G4), launch of BAS-01 and docking mission to BAS-01 (G5).

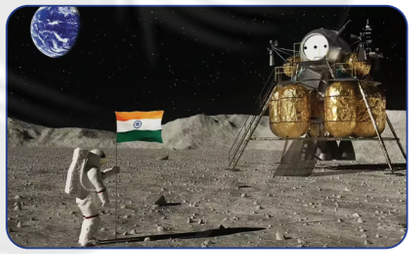
Chandrayaan Programme

- The **Chandrayaan Programme** is India's flagship lunar exploration series managed by ISRO. Starting from a basic orbital map in 2008, the series has evolved into highly complex sample-return and polar exploration missions.

Mission	Launch Vehicle	Launch Date	Landing / Impact Date	Configuration	Key Achievement / Status	Note
Chandrayaan-1	PSLV-C11	Oct 22, 2008	**Nov 14, 2008** (MIP Impact)	Orbiter + Moon Impact Probe (MIP)	**Major Discovery:** Intentionally crashed the MIP at the South Pole to confirm surface water molecules (HO).	Operated for about 310 days; communication lost on 28–29 August 2009

Chandrayaan-2	GSLV-Mk III / LVM3	Jul 22, 2019	**Sep 07, 2019** (Landing Attempt)	Orbiter + Vikram Lander + Pragyan Rover	**Partial Success:** The orbiter remains safely operational in lunar orbit, but the lander crashed during its final descent.	Orbiter mass: ~2,379 kg; Despite failure, its orbiter continues to send data
Chandrayaan-3	GSLV-Mk III / LVM3	Jul 14, 2023	**Aug 23, 2023** (Soft Landing)	Propulsion Module + Vikram Lander + Pragyan Rover	**Historic Success:** Achieved the world's first successful soft-landing near the Lunar South Pole (Shiv Shakti).	**Fourth nation overall** to land on the Moon (US, Russia, China, then India)
Chandrayaan-4	Two **LVM3 (GSLV Mk III)** rockets	2027-28 (Planned)	—	5-Module Stack (2x LVM3 launches)	**Sample Return:** Planned to land at the Mons Mouton region to collect and return lunar soil back to Earth.	Chandrayaan-4 will consist of five modules split into two spacecraft stacks
Chandrayaan-5 (LUPEX)	—	2028-29 (Planned)	—	Joint ISRO-JAXA Lander & Rover	**Ice Analysis:** Polar exploration mission with Japan to drill and analyze subsurface water ice quantity.	JAXA (Japan) – responsible for the launch vehicle (H3-24L) and the lunar rover

Chandrayaan-3

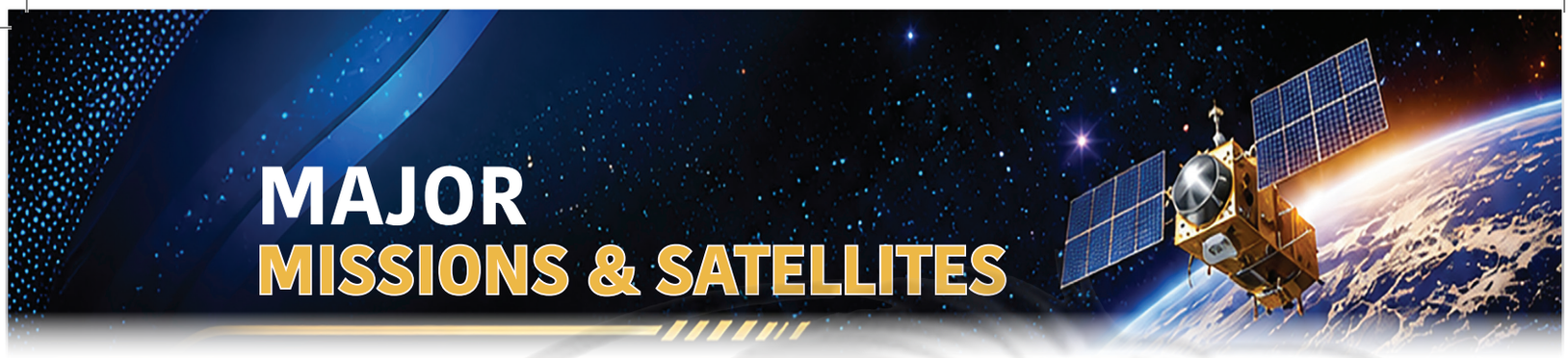


- Chandrayaan-3 was India's historic mission to demonstrate end-to-end capability in safe landing and roving on the lunar surface.
- **Mission Profile:**
  - The LVM3 M4 rocket, nicknamed Bahubali or Fatboy, placed the module into an Elliptic Parking Orbit of 170 by 36500 km.
  - This was the 4th operational flight of India's heaviest and most capable launch vehicle.
  - Launch Date: July 14, 2023.
  - Launch Site: Satish Dhawan Space Centre (SDSC) SHAR, Sriharikota.
  - Landing Date: August 23, 2023.
  - Landing Site: Near the lunar South Pole (69.37°S, 32.32°E).
- **Spacecraft Modules:**
  - Propulsion Module (2,148 kg): Transported the craft to lunar orbit; carried the SHAPE payload.
  - Lander (Vikram - 1,752 kg): Executed a soft landing; featured ChaSTE (thermal) and ILSA (seismic) payloads.
  - Rover (Pragyan - 26 kg): Conducted in-situ soil analysis using LIBS and APXS.
- Mission Goal: Designed for one lunar day (14 Earth days) to demonstrate safe landing, robotic roving, and lunar surface experiments.
- Significance: India became the fourth nation to achieve a soft lunar landing and the first to land near the lunar South Pole, successfully overcoming the challenges faced by Chandrayaan-2.

Shukrayaan-1



- The Venus Orbiter Mission (VOM), colloquially known as Shukrayaan-1, is India's maiden interplanetary mission to Venus.
- The mission was formally approved by the Union Cabinet on September 18, 2024, with a budget of ₹1,236 crore.
- Launch Date: March 29, 2028 (Planned)
- Transit Duration: 112-day journey (reaching Venusian orbit on July 19, 2028)
- Launch Vehicle: Upgraded LVM3 (Launch Vehicle Mark 3)
- Mission Life: 4 years
- Target Orbit: A 200 km × 600 km polar science orbit (achieved gradually from an initial elliptical orbit via aerobraking).



MAJOR MISSIONS & SATELLITES

Mission / System	Description	Key Details / Status
TRISHNA	**Thermal InfraRed Imaging Satellite for High-resolution Natural Resource Assessment.** Jointly developed by ISRO and CNES.	**Launch:** Scheduled in 2027. **Capability:** High-resolution thermal imaging of land/ coastal regions.
NavIC (NDSS)	**India's indigenous satellite navigation system.** Provides accurate positioning, navigation, and timing (PVT) across India and up to 1,500 km beyond.	**Strategy:** Reduces dependence on foreign systems. **Constellation:** 2nd gen satellites (NVS-01 to NVS-05). **Launches:** NVS-01 (May 2023), NVS-02 (Jan 2025). **Adoption:** Integrated in chipsets (e.g., Qualcomm); Ref. station in South Africa (2025 agreement).
GAGAN	**GPS Aided GEO Augmented Navigation.** Satellite-Based Augmentation System (SBAS) by ISRO and AAI.	**Status:** Fully operational since 2015. **Milestone (2026):** DGCA conducted India's first satellite-based landing approach on a commercial jet.
POEM-4	**PSLV Orbital Experimental Module.** Used PSLV's 4th stage (PS4) as an orbital platform.	**Payloads:** 24 total (10 from non-govt, 14 from ISRO). **Milestone:** Completed **1,000 orbits** on March 4, 2025.
CROPS-1	Compact Research Module for Orbital Plant Studies. Unmanned module for growing plants in space.	**Mission:** Flew on POEM-4 (PSLV-C60). **Goal:** Seed germination up to two-leaf stage. **Duration:** 5-7 days in orbit. **Seed:** Cowpea selected after ground tests.
Chang'e 4	China's lunar mission	First nation to land a rover on the **far side** of the Moon (Von Karman crater)

LATEST INFRASTRUCTURE & LAUNCH VEHICLES



Project / Vehicle	Description	Capacity / Status
Third Launch Pad (TLP)	New launch infrastructure at **SDSC SHAR,** Sriharikota.	**Purpose:** Supports **NGLV** and heavy-lift rockets; standby for existing pads. **Investment:** ₹4,000 Crores (approved by Union Cabinet). **Target:** Operational by **2029–30.**

NGLV	**Next Generation Launch Vehicle.** Heavy-lift, partially reusable vehicle.	**Payload:** ~30t to LEO; ~10t to GTO. **Features:** Reusable first stage, semi-cryogenic engines, modular design. **Goal:** Supports Space Station, Gaganyaan, and commercial launches. **Budget:** ~₹8,240 Crores.
SSLV	Small Satellite Launch Vehicle.	**Tech Transfer:** Transferred to **HAL** (Sep 2025). **Launch Complex:** Dedicated complex being established at **Kulasekarapattinam**, Tamil Nadu.
Vikram-1 (Skyroot)	Private launch vehicle by Skyroot Aerospace.	**Test:** First static test of Kalam-1200 motor (Aug 8, 2025).
Agnibaan SOrTeD	Suborbital demonstrator by **Agnikul Cosmos** (IIT Madras spin-off).	**Launch Date:** May 30, 2024. **Place:** Agnikul's private launchpad ("Dhanush") at SDSC-SHAR. **Engine:** "Agnilet" – world's first single-piece, 3D-printed semi-cryogenic engine. **Firsts:** First private Indian launchpad launch; first Linux-based Indian flight.
CE20	Cryogenic engine for LVM3.	**Status:** Key ground test demonstrated higher thrust capability.



TECHNOLOGY & INNOVATION (MAKE-IN-INDIA)

Component	Description	Significance
VIKRAM 3201	First fully Make-in-India **32-bit microprocessor** for launch vehicles.	**Tech:** 180nm CMOS plant (Semi-Conductor Lab). **Upgrade:** Improved version of VIKRAM 1601 (used since 2009). **Env:** Qualified for harsh launch conditions.
KALPANA 3201	32-bit **SPARC V8 RISC microprocessor.**	**Tech:** IEEE 1754 ISA; supports open-source tools. **Test:** Tested with real flight software.
APSARA-U	**Apsara-Upgraded** research reactor by BARC	**Fuel:** First Indian research reactor using **Low Enriched Uranium (LEU).** **Criticality:** Achieved on Sept 10, 2018
Bharat Forecast System (BFS)	India's first high-resolution weather model.	**Resolution:** 6 km x 6 km. Developed by IITM, Pune; operationalised by IMD
PARAS-2	Spectrograph at PRL, Mount Abu Observatory.	**Discovery: Confirmed exoplanet TOI-6038A b (sub-Saturn size, 78.5 Earth masses).**



INTERNATIONAL COOPERATION & EVENTS

Event / Initiative	Details
GLEX 2025	**Global Space Exploration Conference.** **Host:** New Delhi (May 7-9, 2025). **Theme:** "Reaching New Worlds: A Space Exploration Renaissance".
IAC 2025	**76th International Astronautical Congress.** **Host:** Sydney (Sep 29 – Oct 3, 2025). **Activity:** ISRO & IN-SPACe established India Space Pavilion.
SMOPS-2026	**Spacecraft Mission Operations Conference.** **Host:** Bangalore (Apr 8-10, 2026). **Theme:** Innovative Operations for Smart & Sustainable Mission Management.
Future Hosting (2026)	India will host: 1. ICG (International Committee on GNSS). 2. BRICS Heads of Space Agencies. 3. International Planetary Data Alliance.
NASA Exoplanet Milestone	Confirmed **6,000 exoplanets** beyond our solar system

OUTREACH PROGRAMS

NE SPARKS



- The North East Students' Programme for Awareness, Reach, and Knowledge on Space (NE-SPARKS) is a pioneering initiative aimed at igniting curiosity and fostering awareness about space science and technology among students from the North Eastern Region (NER) of India.
- This program seeks to bridge geographical and informational gaps by providing students with an immersive experience of India's advancements in space research and exploration through visits to ISRO Centres in Bengaluru.

UNispace Nanosatellite Assembly & Training by ISRO (UNNATI)



- India announced this programme in 2018 in Vienna during the Symposium to commemorate the 50th Anniversary of the First United Nations Conference on the Exploration and Peaceful Uses of Outer Space (UNISPACE+50).
- ISRO, Department of Space, Govt. of India offers a training programme on nanosatellite assembly and building, UNNATI (UNispace Nanosatellite Assembly & Training by ISRO). Foreign Engineers/Scientists/Officials are eligible to get trained.

Institutional Frameworks by the Government to Encourage Space Start-Ups

1. **2023** – Indian Space Policy
2. **2023** – IN-SPACe Seed Fund Scheme
3. **2024** – Edu-cation IGS Policy
4. **2024** – Norms, Guidelines and Procedures for Authorisation
5. **2024** – IN-SPACe Pre-incubation Programme (IN-SPACe PI) Program
6. **2024** – Establishment of IIC, ISRO–IIT Madras Space Tech Fund
7. **2025** – Establishment of IRL, ISRO Centre Technology Incubation Fund (TIF)
8. **2025** – Technology Transfer of Small Satellite Tech & Hand-holding (SATH) to NFs
9. **2026** – Establishment of IN-SPACe Conservation (IC) & satellite constellation on PPP model

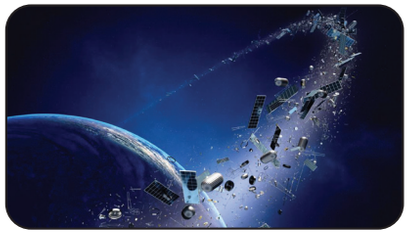
Indian Space Policy 2023

- The Indian Space Policy 2023 serves as a strategic framework designed to institutionalize the reforms initiated in the space sector in 2020.
- The policy clearly delineates the roles of four key entities to prevent conflicts of interest and streamline operations:
  - IN-SPACe (Indian National Space Promotion and Authorisation Centre): Estd. 2020 operational in 2022, it acts as the single-window, autonomous regulatory agency. It authorizes space activities for both government and non-government entities (NGEs) and ensures a level playing field.
  - ISRO (Indian Space Research Organisation): Transitions its focus toward advanced research and development (R&D), cutting-edge technology, and human spaceflight. It will move away from manufacturing operational space systems, transferring mature technologies to the industry.
  - NSIL (NewSpace India Limited): Estd. in 2019, a Public Sector Undertaking (PSU) responsible for the commercialization of space technologies and platforms. It will operate on commercial principles, including manufacturing, leasing, and procuring space assets.
  - Department of Space (DOS): Provides the overall policy guidelines, coordinates global space governance, and manages dispute resolution mechanisms.

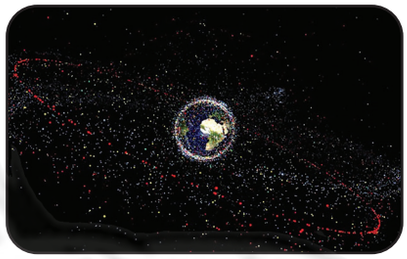


Debris-Free Space Mission (DFSM)

- The Debris-Free Space Mission (DFSM) is a strategic initiative announced by ISRO in 2024, aiming to achieve the goal of zero debris creation for all Indian space missions—both governmental and non-governmental, by the year 2030.



Space Situational Awareness (SSA)



- Space Situational Awareness (SSA) refers to the ability to track, monitor, and predict the behaviour of objects and hazards in space to protect satellites, spacecraft, and ground infrastructure. It is a critical component of modern space programmes for ensuring safety, security, and sustainability of space operations.

2024 Foreign Direct Investment (FDI) Policy Amendment



- The 2024 FDI policy liberalisation and the IN-SPACe Norms, Guidelines, and Procedures (NGP) are key pillars for growing India's space economy:
- Liberalised FDI Policy (2024)
  - 100% Automatic Route: Manufacturing of satellite components, systems, and sub-systems.
  - Up to 74% Automatic Route: Satellite manufacturing, operations, data products, and ground/user segment services. (Beyond 74% requires government approval).
  - Up to 49% Automatic Route: Launch vehicles, associated systems, and spaceports. (Beyond 49% requires government approval).

IN-SPACe Norms, Guidelines and Procedures (NGP) 2024



- Purpose: Provides a predictable regulatory framework to implement the Indian Space Policy 2023.
- Key Function: Acts as a single-window digital portal for private entities (NGEs) to obtain mandatory authorisations for activities like satellite launches, ground station operations, and remote sensing data dissemination.

GEOPORTALS AND CITIZEN-CENTRIC DIGITAL PLATFORMS

- ISRO operates specialised geoportals providing satellite-derived information for governance and public use.
- Bhoonidhi, MOSDAC, VEDAS, and Bhuvan support Earth observation, weather services, infrastructure monitoring, flood management, and digital addressing through DIGIPIN integration.

INTERNATIONAL SPACE

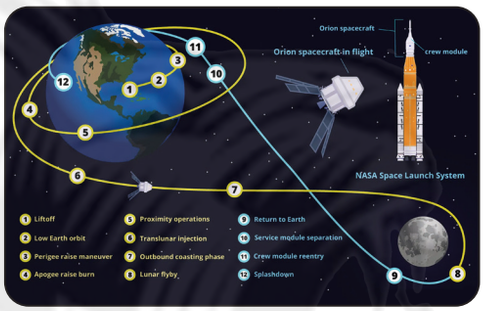
Apollo Missions



- Apollo 11 (16–24 July 1969): First crewed Moon landing. Neil Armstrong and Buzz Aldrin walked on the Moon while Michael Collins stayed in orbit.
- Apollo 12 (14–24 Nov 1969): Precise landing in the Ocean of Storms; extended surface operations and deployments.
- Apollo 13 (11–17 Apr 1970): An oxygen tank explosion crippled the service module; the crew did not land but safely returned to Earth after looping around the Moon.
- Apollo 14 (31 Jan–9 Feb 1971): Landed in the lunar highlands; conducted surveys and sample collection.
- Apollo 15 (26 July–7 Aug 1971): First use of the Lunar Roving Vehicle (LRV); explored near mountainous terrain and collected samples including the "Genesis Rock".

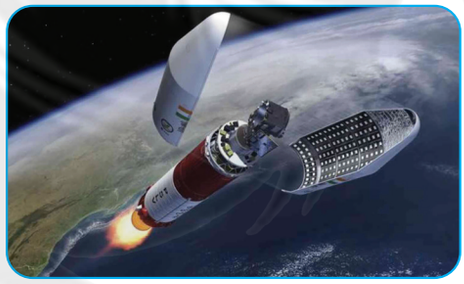
- Apollo 16 (16–27 Apr 1972): Landed in the Descartes region; focused on geological studies and collected large samples such as “Big Muley”.
- Apollo 17 (7–19 Dec 1972): Last Apollo Moon landing; crew spent more time on the surface and drove the LRV farther than any previous mission.

Artemis II Mission



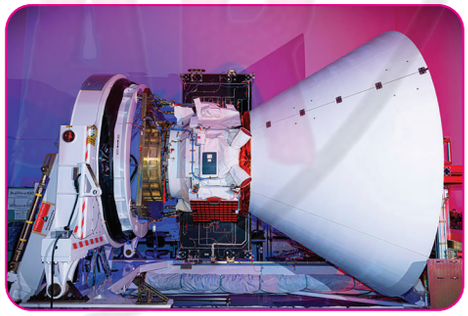
- **Launch Date:** 1 April, 2026
- **Splashdown Date:** 10 April, 2026
- **Duration:** 9 days, 1 hour, and 32 minutes
- **Spacecraft Used:** Orion Crew Module (CM-003, named Integrity by the crew) integrated with the European Service Module (ESM-2)
- **Launch Architecture:** Propelled by the Space Launch System (SLS)
- **Crew:**
  - Reid Wiseman (Commander)
  - Victor Glover (Pilot)
  - Christina Koch (Mission Specialist)
  - Jeremy Hansen (Mission Specialist)

Proba-3 Mission



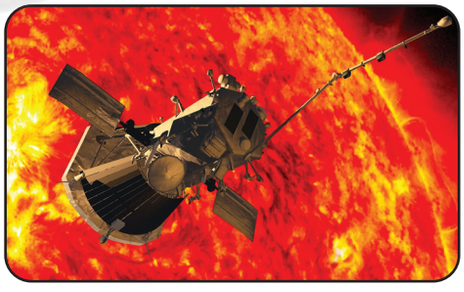
- Launch Vehicle: PSLV-C59
- 4 propulsion stages
- Lift-off mass: 320 tonnes
- Proba-3 is an European Space Agency (ESA) In-Orbit Demonstration (IOD) mission comprising two spacecraft: the Coronagraph Spacecraft (CSC) and the Occulter Spacecraft (OSC), line up to cast a precisely controlled shadow from one to the other, creating an **artificial solar eclipse on demand.**
- It was launched from the First Launch Pad (FLP) at the Satish Dhawan Space Centre (SDSC-SHAR) in Sriharikota.
- **Orbit:** Highly Elliptical Earth Orbit (HEO) with an apogee of approximately 60,500 km.

SPHEREx



- **SPHEREx** (**S**pectro-**P**hotometer for the **H**istory of the Universe, **E**poch of **R**eionization, and **I**ces **E**xplorer) is a NASA astrophysics space telescope launched on **March 11, 2025.**
- It is performing the first-ever near-infrared all-sky spectral survey to create an unprecedented 3D map of the cosmos.
- **Launch Date:** March 11, 2025 (via a SpaceX Falcon 9 rocket)
- **Mission Duration:** 27-month primary mission (capturing a full all-sky map once every 6 months)
- **Orbit:** Sun-synchronous Low Earth Orbit (LEO) at an altitude of 700 km

Parker Solar Probe



- NASA mission to “touch the Sun”; first to enter the corona.
- Launch: 12 August 2018, from Cape Canaveral, on a Delta IV Heavy rocket.
- Closest distance: ~3.8 million miles from the Sun’s surface.
- Speed: ~430,000 mph – fastest human-made object.
- Named after: Eugene N. Parker, who predicted the solar wind.

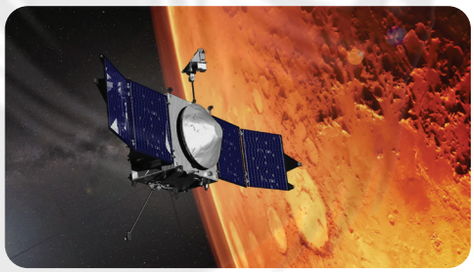
- Main goals: Study solar corona heating, solar wind acceleration, and energetic particles.
- Importance: Helps understand and forecast space weather that affects Earth's technological systems.

OSIRIS-REx



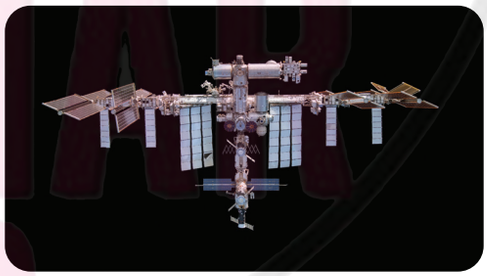
- NASA's first U.S. asteroid sample-return mission.
- Launch: 8 September 2016; returned sample on 24 September 2023.
- Target: Near-Earth asteroid Bennu (carbon-rich, potentially hazardous).
- Sample mass: ~250 g; largest extraterrestrial sample since Apollo.
- Key findings:
  - Bio-essential sugars.
  - Unique organic "gum-like" substance.
  - High levels of supernova dust.
- Follow-on mission: Spacecraft renamed OSIRIS-APEX, will study Apophis in 2029.
- Significance: Clues about origin of life, water delivery, and asteroid impact risks.

MAVEN (Mars Atmosphere and Volatile EvolutioN)



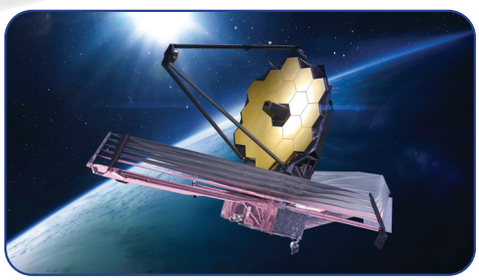
- **MAVEN** was a pioneering NASA orbiter dedicated to studying the Martian upper atmosphere.
- **Launch:** November 18, 2013, from Cape Canaveral.
- **Arrival at Mars:** September 21, 2014.
- **Mission Duration:** Originally designed for a two-year mission, it successfully operated for over 11 years, far exceeding its planned lifespan.
- **Discovery of Auroras:** Identified new types of Martian auroras caused by solar energetic particles.
- **Relay Service:** Served as a critical high-speed communications relay for NASA's surface missions, including the **Curiosity** and **Perseverance** rovers.
- **End of Mission:** On **June 3, 2026,** NASA officially declared the MAVEN mission concluded after declaring the spacecraft unrecoverable.

International Space Station (ISS)



- **Partnership:** It is operated by five space agencies: **NASA** (USA), **Roscosmos** (Russia), **ESA** (Europe), **JAXA** (Japan), and **CSA** (Canada).
- **First launch:** First module launched in **1998;** first crew arrived in **2000;** continuously inhabited since then.
- **Orbit:** It travels at approximately 28,000 km/h at an altitude of about 400–430 km, completing an orbit of Earth every 90 minutes.
- The normal crew size is **seven people,** though it can be higher during crew handovers.
- ISS is a habitable artificial satellite (space station) in **Low Earth Orbit (LEO).**
- **Future Outlook:** The station is being utilized for scientific research while partners discuss its transition toward the 2030s.

James Webb Space Telescope (JWST)



- JWST is the largest and most powerful space telescope ever built, optimized for infrared observations.
- It works at the L2 point, 1.5 million km from Earth, not in Earth orbit.
- It is a joint mission of NASA, ESA, and CSA.

- Main goals: study first galaxies, star/planet formation, and exoplanet atmospheres.
- Launched on 25 December 2021, first major image released on 11 July 2022.

Event Horizon Telescope (EHT)



- The **Event Horizon Telescope (EHT)** is an international array of globally distributed radio telescopes designed to act as a single, Earth-sized virtual telescope.
- **M87 (2019):** The EHT released the first-ever image of a black hole's shadow in the galaxy Messier 87. It confirmed the existence of the "photon ring" and the silhouette predicted by Einstein's General Relativity.
- **Sagittarius A (2022):** The EHT imaged the supermassive black hole at the center of our own Milky Way galaxy.
- **Spin and Magnetic Fields (2024):** The EHT captured polarized light images of M87, revealing the powerful, organized magnetic fields that allow black holes to launch relativistic jets of matter across thousands of light-years.

LIST OF FIRSTS

1. First artificial satellites and space probes

- First artificial Earth satellite: Sputnik 1 (Soviet Union), launched on 4 October 1957 – the first object to orbit Earth intentionally.
- First animal to orbit Earth: Laika (dog), aboard Sputnik 2, launched 3 November 1957 – first animal in orbit (though she died during the mission).
- First US satellite: Explorer 1, launched 31 January 1958 – carried instruments that discovered Earth's radiation belts.
- First spacecraft to escape Earth's gravity and reach the Sun: Luna 1 (Soviet), 2 January 1959 – first to achieve Earth escape velocity and fly past the Moon.
- First spacecraft to impact another planet: Venera 3 (Soviet), 16 November 1965 launch, impacted Venus on 1 March 1966 – first hard impact on another planet.
- First soft landing on the Moon: Luna 9 (Soviet), launched on 31 January, 1966 and landed 3 Feb 1966 – first successful lunar soft landing.
- First US soft landing on the Moon: Surveyor 1, landed 2 June 1966 – returned televised images of the lunar surface.
- First spacecraft to orbit the Moon: Luna 10 (Soviet), Launch – 31 March 1966 – first to be placed in sustained lunar orbit.

2. First humans in space

- First human in space: Yuri Gagarin (Soviet), Vostok 1, 12 April 1961 – single-orbit flight around Earth.
- First American in space: Alan Shepard, May 5, 1961 – a suborbital flight (not orbit).
- First American in Earth orbit: John Glenn, Mercury-Atlas 6, 20 February 1962 – first US orbital flight.
- First woman in space: Valentina Tereshkova (Soviet), Vostok 6, 16–19 June 1963.
- First multi-person spacecraft: Voskhod 1 (Soviet), 12 October 1964 – carried three cosmonauts.
- First US multi-person spacecraft: Gemini 3, 23 March 1965 – carried two astronauts.

3. First space manoeuvres and operations

- First spacewalk (egress from spacecraft): Aleksei Leonov (Soviet), Voskhod 2, 18 March 1965 – floated outside the spacecraft for ~12 minutes.
- First American spacewalk: Ed White, Gemini 4, 3 June 1965.
- First orbital rendezvous: Gemini 6 and Gemini 7, 15 December 1965 – first time two spacecraft met in orbit.
- First docking in space: Gemini 8 docking with an Agena target vehicle, 16 March 1966 – first spacecraft-to-spacecraft docking.
- First human flight around the Moon and return: Uncrewed Soviet probe Zond 5, 14–21 September 1968 – flew around the Moon and returned to Earth.

- First crewed flight around the Moon: Apollo 8 (US), 21–27 December 1968 – first humans to orbit the Moon and see Earthrise.
- First international docking in space (US–Soviet): Apollo–Soyuz Test Project (ASTP), 15 July, Docking 17 July, 1975 – first joint US–Soviet mission; docked Apollo and Soyuz spacecraft.

4. First Moon landings and surface operations

- First humans to land on the Moon: Apollo 11 (US), 20–21 July 1969 – Neil Armstrong and Buzz Aldrin walked on the Moon while Michael Collins orbited.
- First person to walk on the Moon: Neil Armstrong, Apollo 11, 20 July 1969 – first human steps on lunar surface.
- First robotic rover on the Moon: Soviet Lunokhod 1, 1970 – first unmanned lunar rover.
- First woman to perform a spacewalk: Svetlana Savitskaya (Soviet), 1984.
- First American woman in space: Sally Ride, 1983 (Space Shuttle).

5. First space stations and long-duration occupation

- First space station: Salyut 1 (Soviet), launched 1971 – first artificial orbital habitat for humans.
- First long-duration inhabited space station: Mir (Soviet/Russia), launched 1986, continuously occupied from 1989–2000 – record for continuous human presence before ISS.
- First continuously inhabited space station (modern era): International Space Station (ISS) – first crew arrived 22 November 2000, continuous habitation since then.
- First privately funded human spaceflight to a space station: Tourist flight to Mir in 2001 (funded by MirCorp).
- First space tourist: Dennis Tito, 2001 – flew to ISS as a private passenger.
- First privately developed manned spacecraft: SpaceShipOne, 2004 – first private manned spacecraft flight; pilot Mike Melvill.

6. First deep-space and planetary exploration milestones

- First spacecraft to explore the asteroid belt and fly by Jupiter: Pioneer 10, launched 3 March 1972 – first to cross the asteroid belt, fly by Jupiter, and eventually leave the solar system.
- First spacecraft to land on Mars: Viking 1 (US), 20 July 1976 – first successful Mars landing and surface operation.
- First human-made object to leave the solar system: Pioneer 10, later Pioneer 11, Voyager 1, and Voyager 2 all crossed into interstellar space; Voyager 1 is the farthest and first to officially enter interstellar space (2012).

7. First major telescopes and observatories in space

- First major orbiting space telescope: Hubble Space Telescope, launched 1990 – revolutionized astronomy with deep-space images and precise measurements.
- Other landmark missions:
  - Chandra (X-rays), Spitzer (infrared), Kepler (planet search), James Webb (infrared, launched 2021) – expanded our view of the universe across wavelengths.

IMPORTANT FIRMS TO REMEMBER

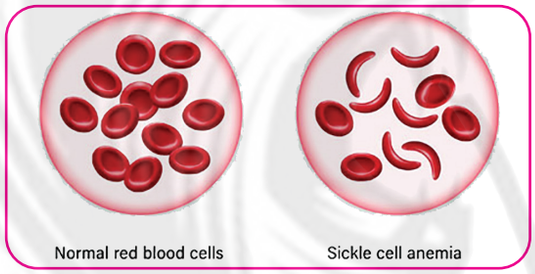
- **Skyroot Aerospace:** Vikram rocket family; first private Indian sub-orbital launch.
- **Agnikul Cosmos:** Agnibaan launch vehicle; 3D-printed rocket-engine capability.
- **Pixxel:** Hyperspectral Earth-observation satellite constellation.
- **Dhruva Space:** Satellite platforms, deployers and space-grade systems.
- **Digantara:** Space situational awareness and space-surveillance services.
- **Bellatrix Aerospace:** Electric propulsion and orbital-transfer technologies.
- **GalaxEye:** Radar-optical Earth-observation data fusion.
- **SatSure:** Satellite-data analytics for agriculture, finance and climate applications.



BIOTECHNOLGY

SCREENING SICKLE CELL ANAEMIA

- **CSIR-CCMB** (Centre for Cellular & Molecular Biology) has developed an accurate, rapid, and affordable first-of-its-kind molecular test for screening sickle cell anaemia (SCA).
- This innovation is a significant step towards combating the disease, particularly in underserved regions where SCA prevalence is high, at a cost of less than Rs 100.



GENOME INDIA PROJECT

- It was initiated in **2020** by the **Department of Biotechnology (DBT)**, Government of India with collaborations of 20 institutions to map India's genetic diversity.
- On **9th January 2025,** Union Minister Dr. Jitendra Singh launched the 'Framework for Exchange of Data Protocols (FeED) and Indian Biological Data Centre (IBDC), making **10,000 whole genome samples** accessible to global researchers.



ZYCOV-D AND CORBEVAX

- The **National Biopharma Mission (NBM)** has developed two COVID-19 vaccines - ZyCoV-D and Corbevax.
- It also created important medicines like **Liraglutide** for diabetes and **pegylated interferon alpha-2b** for COVID-19.
- NBM further produced **India's first MRI scanner,** bioreactors, COVID-19 test kits, ventilators, and lab reagents.



GENOME-EDITED RICE

- ICAR has developed India's first genome-edited rice varieties – **DRR Rice 100 (Kamla)** and **Pusa DST Rice 1.**
- Developed using genome-editing technology based on **CRISPR-Cas.**
- **DRR Rice 100 (Kamala)** variety was developed by **ICAR-IIRR, Hyderabad,** based on Samba Mahsuri (BPT 5204).
- **Pusa DST Rice 1**, was developed by **ICAR-IARI, New Delhi,** based on MTU 1010.



SRJANAM

- It is India's first fully automated biomedical-waste treatment rig, developed by the CSIR–National Institute for Interdisciplinary Science and Technology (CSIR-NIIST), Thiruvananthapuram.
- It was launched at AIIMS New Delhi in February 2025 by Union Minister Dr Jitendra Singh.
- India's first Animal Stem Cell Biobank is at BRIC–NIAB, Hyderabad, Telangana; it supports regenerative medicine and cellular therapies for livestock.



DIGITAL/AI/ROBOTICS AND INNOVATION

INDIA AI IMPACT SUMMIT 2026



The **India AI Impact Summit 2026** was held from 19–20 February 2026 in New Delhi.

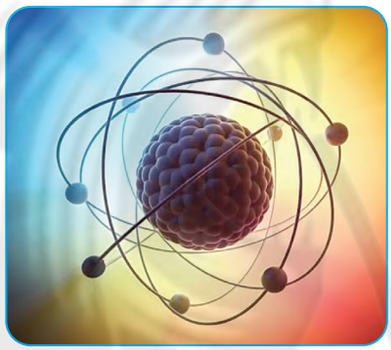
- The Summit was anchored in three pillars: **People, Planet and Progress**
- India formally joined the **Pax Silica coalition** with the USA, strengthening strategic technology and supply chain cooperation.
- A key announcement at the Summit was the expansion of India's sovereign computer capacity.
  - In addition to the 38,000+ GPUs already provisioned under the IndiaAI Mission, an additional 20,000 GPUs will be added in the coming weeks, further strengthening national AI infrastructure.
- The Casebook on AI and Gender Empowerment was officially launched. (jointly by MeitY, UN Women, and supported by MoWCD).
- The Atal Innovation Mission (AIM), under NITI Aayog, launched **"AI-Preneurs of India"**, a flagship coffee table book.
- Union Minister for Health and Family Welfare, Shri Jagat Prakash Nadda, launched two pioneering digital health initiatives
  - **SAHI** (Strategy for Artificial Intelligence in Healthcare for India)
  - **BODH** (Benchmarking Open Data Platform for Health AI)
- Minister of State for Skill Development and Entrepreneurship Jayant Chaudhary launched a nationwide campaign titled **'Badhna Hai Toh Yahan Judna Hai'** to accelerate India's digital skilling movement through Skill India Digital Hub (SIDH).
- **Google** has announced that it will connect India with the US through new subsea cables as part of a 15 billion dollar investment spread over five years.
- **Artificial Intelligence (AI) Impact Summit 2027** will be held at **Geneva, Switzerland.**

MAJOR CORPORATE COMMITMENTS

Company / Entity	Commitment
Reliance Industries	**USD 110 billion** over 7 years
Tata Group	Partnership with **OpenAI**

Adani Enterprises	**USD 100 billion** by 2035
General Catalyst	**USD 5 billion** over 5 years
Lightspeed Venture Partners	USD 10 billion
Sundar Pichai / Google	**USD 15 billion** AI hub in **Visakhapatnam** (India–America Connect Initiative)

NATIONAL QUANTUM MISSION (NQM)



- Approved on 19th April 2023 by the Union Cabinet with an outlay of **Rs. 6,003.65 crore** for a duration of **eight years** (2023–24 to 2030–31).
- **4 Thematic Hubs** have been set up at IISc. Bengaluru, IIT Madras, IIT Bombay and IIT Delhi.
- **8 startups have been supported** - QuNu Labs, QpiAI, Dimira Technologies, PRENISHQ, QuPrayog, Pristine Diamonds, Quanastra, and Quan2D Technologies.
- Three national-level **Central Fabrication Facilities** are established at premier institutions - IIT Delhi, IISc Bengaluru, IIT Bombay and IIT Kanpur with an investment of about **Rs.720 Crore.**

NATIONAL SUPERCOMPUTING MISSION (NSM)



- Launched in **April 2015**
- Outlay - ₹4,500 crore.
- Implementation - DST and Ministry of Electronics and Information Technology (MeitY).
- As of now, supercomputers with a combined capacity of 39 petaflops have been established across 37 locations, up from **34 supercomputers in March 2025.**
- In parallel, under the NSM, **C-DAC** has developed the indigenous high-speed communication network, **"Trinetra,"** to enhance data transfer and communication between computing nodes, strengthening India's supercomputing capabilities.

INNOVATION IN SCIENCE PURSUIT FOR INSPIRED RESEARCH (INSPIRE)

- Department of Science & Technology
- Goal: Attract Young Talent to opt research as a career

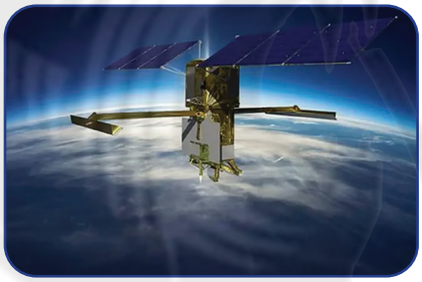
Components:

- INSPIRE INTERNSHIP: aims to provide exposure to the top 1% students at Class X Board level.
- SCHOLARSHIP for HIGHER EDUCATION (SHE): To encourage meritorious students to study B.Sc. & M.Sc. (basic and natural sciences) by providing scholarship (Rs. 60,000/- p.a.) and mentorship grant (Rs. 20,000/- p.a.)
- INSPIRE FELLOWSHIP : It offers 1000 Fellowships every year to bright students in the age group of 22-27 years to pursue Ph.D. in basic and applied sciences including engineering, medicine, agriculture and veterinary sciences.
- INSPIRE FACULTY FELLOWSHIP: the component provides opportunities to 100 persons every year with Ph.D. qualification in the age group of 27-32 years for 5 years to carry out research.
- INSPIRE Faculty Fellowship provides opportunities to 150 persons every year (earlier 100), for Ph.D. holders in the age group of 27–32 years, for 5 years of independent research.

WISE - KIRAN

- 'Women in Science and Engineering-KIRAN (WISE-KIRAN)' aims to **encourage and promote women** in the Research & Development (R&D) in Science & Technology (S&T) domain.
- Core WISE-KIRAN programs
  - **WISE-PhD:** supports women pursuing doctoral research in basic and applied sciences.
  - **WISE-PDF:** supports women with Ph.D. for post-doctoral research in basic and applied sciences.
  - **WISE-SCOPE:** supports translational, lab-to-land research addressing societal challenges.
  - **WISE-IPR:** provides training in Intellectual Property Rights for women.

NATIONAL GEOSPATIAL MISSION



- Announced in the **Budget 2025-26** and **Rs. 100 crores** allocated for the FY 2025-26.
- This mission aims to develop foundational geospatial infrastructure and data.
- **Goals:**
- **By 2030** - Establish a Geospatial Knowledge Infrastructure (GKI) underpinned by an Integrated Data and Information Framework.
- **By 2035** - Develop a National Digital Twin for major urban centers, creating digital replicas to improve urban planning and management.

RASHTRIYA VIGYAN PURASKAR 2025



- **Vigyan Ratna:** Prof. Jayant Vishnu Narlikar (Physics) Posthumously
- **Vigyan Shri:**
  - Dr. Gyanendra Pratap Singh (Agricultural Science);
  - Dr. Yusuf Mohammad Sheikh (Atomic Energy);
  - **Dr. K Thangaraj (Biological Sciences);**
  - Prof. Pradeep Thalappil (Chemistry);
  - Prof. Aniruddha Bhalchandra Pandit (Engineering Sciences);
  - **Dr. S Venkata Mohan (Environmental Science);**
  - Prof. Mahan Mj (Mathematics and Computer Science);
  - Shri Jayan N (Space Science and Technology).
- **CSIR's Aroma Mission** - the team has been selected for the prestigious **Rashtriya Vigyan Puraskar - Vigyan Team Award 2025** under Agricultural Sciences discipline.

The **9th Global Technology Summit** was inaugurated in **New Delhi** by EAM Dr. S. Jaishankar.

Theme: **Sambhavna (Possibilities)**

Transforming India with AI

- 6 million people are employed in the tech and AI ecosystem.
- The tech sector is projected to cross $280 billion in revenue this year.
- AI could add $1.7 trillion to India's economy by 2035.
- **BharatGen AI:** Launched on 2 June 2025 at the BharatGen Summit, it is the first government-funded, homegrown multimodal large language model.
- **India** will host the **AI Impact Summit** in February 2026.

INDIA AI MISSION

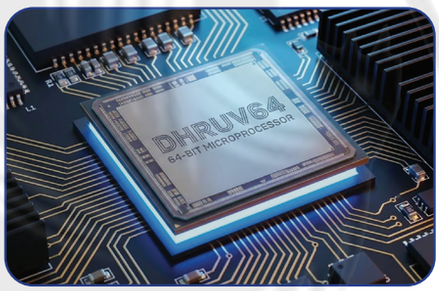
- Cabinet approved the IndiaAI Mission in **March 2024,** with a budget outlay of **₹10,371.92 crore** over five years.
- **Initial Target -** 10,000 GPUs (GPUs, India has now achieved 38,000 GPUs)

PROJECT NETRA

- Project NETRA **(Network for Space Object Tracking and Analysis)** is an indigenous Space Situational Awareness (SSA) initiative launched by the Indian Space Research Organisation (ISRO).
- It acts as an early warning system in space to detect, track, and catalog space debris, inactive satellites, and other orbital hazards, ultimately safeguarding India's critical space assets.
- The sanctioned cost of NETRA project is Rs. 509.01 crore.

DHRUV64

- India launched DHRUV64, its first indigenous 1.0 GHz 64-bit dual-core RISC-V microprocessor
- Developed by C-DAC under MeitY's Microprocessor Development Programme
- 💡 Previous Microprocessors:
  - SHAKTI (2018, IIT Madras)
  - AJIT (2018, IIT Bombay)
  - VIKRAM (2025, ISRO–SCL)
  - THEJAS64 (2025, C-DAC)



9TH INDIA MOBILE CONGRESS

- Inaugurated by PM Modi in New Delhi
- **Theme:** **Innovate to Transform**

